

xBxBio Regulatory-by-Design Framework for Connected Cardiovascular Intelligence

Public Governance Summary Following Internal Review and Approval

Internal review and approval are complete. xBxBio is advancing from framework definition into controlled implementation and validation readiness.

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IMPORTANT: This document describes an internal xBxBio governance milestone. It is not FDA clearance, FDA approval, regulatory authorization, certification, conformity assessment, or external endorsement.

CONTROL

Document Control and Public-Release Boundaries

Document Control

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Public-release boundary

This document summarizes selected non-proprietary governance principles. It does not reproduce the underlying internal framework, controlled procedures, proprietary architecture, source code, model parameters, security-sensitive configurations, trade secrets, unpublished patent material, or confidential implementation methods.

Required interpretation

References to FDA regulations, FDA guidance, ISO standards, IEC standards, IMDRF principles, HL7 FHIR, DICOM, or other external sources identify relevant considerations and reference baselines. They do not establish that xBxBio is certified, compliant, cleared, approved, authorized, validated, or endorsed.

How to Use This Document

This public summary is intended for investors, clinicians, researchers, engineering and data professionals, regulatory and quality professionals, prospective collaborators, and other stakeholders who need a precise description of what xBxBio has completed internally, what phase begins next, and what remains unfinished.

The summary uses forward-looking terms such as “intended,” “planned,” “may,” “should,” and “future.” Those terms are deliberate. They distinguish a governance framework and development direction from a clinically validated, regulator-authorized, publicly deployed product.

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Appendix C. Publication Metadata	Copy-ready Zenodo and GitHub metadata for this document.

EXECUTIVE

Internal Approval Complete — Next Phase Beginning

Executive Statement

xBxBio has completed the internal review and approval of its Regulatory-by-Design Framework for Connected Cardiovascular Intelligence.

Approved public statement

Internal review and approval are complete. xBxBio is now advancing from framework definition into controlled implementation and validation readiness.

The milestone moves xBxBio beyond framework definition and into the phase where regulatory, quality, clinical, technical, data-integrity, cybersecurity, human-factors, and lifecycle principles must be translated into controlled requirements, accountable processes, traceable development records, and predefined evidence expectations.

The next phase includes controlled implementation planning; intended-use and claims definition; design and development controls; risk-management activities; verification and validation planning; data-integrity and provenance controls; cybersecurity requirements; human-factors considerations; clinical AI governance; configuration and change management; and lifecycle-monitoring processes.

This phase is intended to establish the documented controls and evidence architecture needed to support future production development, external evaluation, institutional collaboration, and any applicable regulatory pathway. It does not establish that those later activities have already been completed.

Mandatory regulatory clarification

This represents an internal xBxBio governance milestone. It does not constitute FDA clearance, FDA approval, regulatory authorization, certification, conformity assessment, or endorsement by any governmental agency, notified body, standards organization, healthcare institution, academic institution, or other third party.

The xBxBio Connected Cardiovascular Intelligence Platform remains in pre-commercial research and engineering development. Production cloud deployment, completed formal verification and validation, external clinical evaluation, institutional deployment, regulatory submission, and commercial availability remain future activities dependent on financing, technical execution, appropriate data access, qualified collaborators, evidence development, and applicable regulatory requirements.

1

Purpose, Scope and Intended Audience

1. Purpose, Scope and Intended Audience

1.1 Purpose

The purpose of this document is to describe, at a public and non-proprietary level, the governance foundation that xBxBio has internally reviewed and approved for future development of Connected Cardiovascular Intelligence. It explains the framework's governing logic, the controls that must be operationalized next, the evidence boundaries that apply today, and the prerequisites for future clinical or commercial deployment.

1.2 Scope

The framework addresses regulatory strategy, quality management, design and development controls, risk management, software lifecycle governance, data integrity, provenance, interoperability, clinical AI governance, scientific and clinical validation, human factors, cybersecurity, privacy, change control, post-deployment monitoring, and responsible public communication.

1.3 Exclusions

- This document is not a regulatory submission, pre-submission, quality manual, design history file, risk-management file, software development plan, clinical evaluation report, cybersecurity plan, validation protocol, or certification package.
- It does not establish a final intended use, device classification, regulatory pathway, reimbursement status, clinical claim, labeling claim, deployment claim, or performance claim.
- It does not disclose confidential architecture, source code, model weights, model parameters, training data, security controls, trade secrets, unpublished patent content, or proprietary implementation details.
- It does not authorize use of the xBxBio platform for diagnosis, treatment, patient management, clinical decision-making, or other clinical purposes.

1.4 Intended Audience

The intended audience includes sophisticated external stakeholders who need an accurate, reviewable description of xBxBio's governance maturity without receiving confidential internal documents. This includes prospective investors, strategic partners, cardiovascular clinicians and researchers, software and data engineers, regulatory and quality specialists, cybersecurity professionals, and institutional decision-makers.

2

Milestone and Current Development State

2. Milestone and Current Development State

The milestone is significant because it separates three different states that are often incorrectly collapsed into one: an internally approved governance framework, an implemented and validated production system, and an externally authorized clinical product. xBxBio has completed the first state. The second and third remain future work.

Area	Status	What it means	Public interpretation
Internal regulatory framework	Complete	Internally reviewed and approved governance foundation.	Internal milestone only.

Area	Status	What it means	Public interpretation
Controlled implementation planning	Beginning	Framework principles converted into controlled requirements, roles and records.	Not production deployment.
Production cloud environment	Not complete	Production infrastructure, security, operations and controlled deployment remain future work.	Not publicly available.
Formal platform V&V	Not complete	Protocol-driven verification and validation of the production configuration remain future work.	No validated performance claim.
External clinical evaluation	Not complete	Independent retrospective or prospective evaluation remains future work.	No external clinical validation claim.
Regulatory authorization	Not obtained	Any applicable pathway depends on final intended use and claims.	Not FDA cleared or approved.
Commercial availability	Not available	No public clinical or commercial deployment at this time.	Do not solicit patient use.

2.1 Development Progression

Stage	Phase	Meaning
1	Architecture	Conceptual and technical architecture defined.
2	Internal Governance Approval	Regulatory-by-Design framework internally reviewed and approved.
3	Controlled Implementation	Requirements, procedures, records, roles and controls operationalized.
4	Evidence Development	Engineering, analytical, human-factors and clinical evidence generated.
5	Verification and Validation	Predefined acceptance criteria tested against the production configuration.
6	External Evaluation	Qualified institutions or collaborators independently evaluate appropriate functions.
7	Applicable Regulatory Pathway	Submission or other regulatory activity undertaken when required.
8	Deployment	Controlled clinical or commercial deployment only after prerequisites are satisfied.

3. Regulatory-by-Design Principles

- **Intended use before claims:** The final intended use, users, use environment, patient population, input data, output, clinical action, and limitations must be defined before a defensible regulatory strategy or validation plan can be completed.
- **Quality before scale:** Controlled processes, accountable ownership, traceability, and evidence requirements must precede broad deployment.
- **Risk throughout the lifecycle:** Risk management begins during concept definition and continues through design, development, validation, deployment, monitoring, change, and retirement.
- **Evidence proportional to consequence:** The evidence burden should increase with the potential clinical consequence, degree of automation, uncertainty, irreversibility, and difficulty of detecting error.
- **Provenance before inference:** A derived result should not be treated as reliable if the system cannot identify its source, timing, acquisition context, transformations, version, and validation state.
- **Uncertainty must remain visible:** Uncertainty, missingness, conflict, shift, and evidence insufficiency should be represented rather than suppressed by a single confident output.
- **Human oversight must be engineered:** Human review is not a disclaimer. The interface, workflow, escalation path, decision rights, and training must make oversight operational and testable.
- **Physiology constrains computation:** Cardiovascular models should be evaluated against known physiological relationships and should not be accepted solely because they optimize a statistical metric.
- **Security and privacy are safety properties:** Cybersecurity, identity, access, integrity, availability, confidentiality, resilience, and recovery must be integrated into system design and risk management.
- **Change must be controlled:** Software, models, data sources, interfaces, dependencies, thresholds, labels, and clinical claims require traceable impact assessment and controlled release.
- **Public statements must reflect evidence:** Marketing, investor, scientific, and press statements must not exceed the actual development, validation, authorization, and deployment status.

4. Governance, Accountability and Decision Rights

A governance framework becomes operational only when decision rights, accountability, independence, escalation, and evidence ownership are assigned. Before controlled implementation begins, xBxBio should establish and document the following roles. A single qualified person may hold more than one role during an early stage, but incompatible review and approval functions should be separated where risk warrants independent review.

Governance role	Primary accountability	Minimum decision authority
Accountable executive management	Resources, quality policy, management responsibility and risk acceptance.	Approve phase gates, material residual risk and public status statements.
Regulatory and quality lead	Regulatory strategy, QMS structure, controlled documents, design controls and audit readiness.	Approve regulatory classifications, quality records and release gates.
Clinical safety lead	Clinical context, foreseeable harm, workflow safety, escalation and evidence sufficiency.	Stop or restrict development or release when clinical risk is unacceptable.
Software and systems owner	Architecture, requirements, configuration, software lifecycle, integration and release.	Approve technical baselines and implementation changes.

Governance role	Primary accountability	Minimum decision authority
Data governance and provenance owner	Data rights, quality, lineage, identity, semantics, retention and access.	Accept or reject data sources and transformations.
AI/model governance owner	Model purpose, training/validation boundaries, performance, uncertainty, drift and change.	Approve model baselines, thresholds and monitoring plans.
Cybersecurity and privacy lead	Threat modeling, security architecture, privacy controls, incident response and resilience.	Approve security risk controls and release readiness.
Independent reviewer(s)	Objective review of evidence, risk, verification, validation or claims.	Recommend approval, rejection, remediation or additional evidence.

4.1 Required Decision Gates

- Concept and intended-use gate
- Regulatory-classification and claims gate
- Architecture and data-governance gate
- Risk-control and design-input approval gate
- Verification-readiness gate
- Validation-readiness gate
- External-evaluation readiness gate
- Production-release and deployment gate
- Change-release and rollback gate
- Public-communications approval gate

5

Intended Use and Regulatory Strategy

5. Intended Use and Regulatory Strategy

The regulatory status of future xBxBio functionality cannot be determined from the company name or architecture alone. It will depend on the final intended use, claims, users, patient population, input data, output, degree of automation, clinical consequence, deployment context, and jurisdiction. The framework therefore requires intended-use definition and claims control before product-level regulatory conclusions are made.

In the United States, the current [FDA Clinical Decision Support Software guidance](#) and the FDA's broader [digital-health guidance library](#) provide relevant reference points for evaluating software functions. These sources must be applied to the final, fact-specific functionality; citation alone does not establish a regulatory conclusion.

5.1 Minimum Intended-Use Definition

- Who is the intended user, and what qualifications or training are required?
- What patient population, condition, clinical state, or research context is addressed?
- What data enter the function, and what assumptions govern their quality and availability?
- What output is produced, how is it communicated, and what action may follow?
- Does the user independently review the basis of the recommendation or inference?
- What is the foreseeable consequence of an incorrect, delayed, omitted, or misunderstood output?
- Is the function advisory, prioritizing, monitoring, predictive, diagnostic, therapeutic, autonomous, administrative, or research-only?
- What limitations, contraindications, exclusions, and abstention conditions apply?

5.2 Claims Registry

A controlled claims registry should be created before the next public product statement. Each proposed claim should identify the exact wording, audience, channel, intended use, supporting evidence, platform version, responsible owner, approval date, limitations, and review date. Claims should not be published merely because a prototype appears capable of producing a result.

6

Quality Management and Design Controls

6. Quality Management and Design Controls

The [FDA Quality Management System Regulation \(QMSR\)](#) became effective on February 2, 2026 and incorporates ISO 13485:2016 by reference for applicable finished-device manufacturers. xBxBio uses the QMSR and [ISO 13485:2016](#) as important regulatory and quality reference points; this public document does not claim that xBxBio is currently subject to every provision, has completed implementation, or is certified.

6.1 Controlled Quality-System Elements for the Next Phase

- Quality policy, measurable quality objectives, management responsibility and management review.
- Document and record control, approval authority, effective dates, revision history and retention.
- Competence, training, role qualification and independent-review requirements.
- Design and development planning, stage gates, design reviews and controlled baselines.
- User needs, intended use, system requirements, software requirements and traceability.
- Design outputs, architecture specifications, interface specifications and acceptance criteria.
- Verification, validation, usability, cybersecurity, data and model evaluation records.
- Supplier, cloud, model, data, software-component and service-provider controls.
- Nonconformity, corrective and preventive action, complaint and feedback handling as applicable.
- Release, deployment, change control, monitoring, incident response and retirement controls.

6.2 Design Traceability

Each safety- or performance-relevant claim should trace from intended use and user need through design inputs, architecture and design outputs, implemented components, identified risks, verification methods, validation evidence, labeling or communication, and lifecycle monitoring. Traceability should include data and model versions, not software requirements alone.

7

Risk Management and Clinical Safety

7. Risk Management and Clinical Safety

[ISO 14971:2019](#) provides a lifecycle framework for medical-device risk management, supported by [ISO/TR 24971:2020](#). The newly published [ISO/TS 24971-2:2026](#) provides guidance for applying ISO 14971 to machine-learning-enabled medical devices within its defined scope.

Important scope limitation

ISO/TS 24971-2:2026 expressly excludes ML-enabled medical devices employing large language models or generative AI. It therefore cannot be treated as the sole AI-risk reference for any future xBxBio generative-AI or LLM function. Broader AI governance and risk references, including ISO/IEC 23894 and ISO/IEC 42001, may also be relevant depending on the final architecture and intended use.

7.1 Representative Risk Categories

- **Incorrect inference:** False-positive, false-negative, misclassification, incorrect severity, incorrect trajectory or unsupported causal attribution.
- **Omitted or delayed information:** A clinically relevant change is not detected, communicated or escalated within the required time.
- **Data-integrity failure:** Wrong patient, wrong encounter, wrong timestamp, unit conversion error, corrupted data, duplicate data or incomplete provenance.
- **Distributional change:** Performance degrades across institutions, devices, populations, workflows, acquisition protocols or time.
- **Automation bias:** A user over-relies on an output or underweights contradictory clinical evidence.
- **Interaction failure:** A correct output is misunderstood because of interface, workflow, terminology, prioritization or alert design.
- **Model or software change:** A modification alters performance, calibration, subgroup behavior, cybersecurity or clinical interpretation.
- **Cybersecurity event:** Unauthorized access, tampering, denial of service, data exfiltration, dependency compromise or loss of availability.
- **Unverifiable provenance:** A derived output cannot be reconstructed from its source data, transformations, model version and assumptions.
- **Research-to-clinical leakage:** Exploratory findings are presented or used as though they were validated clinical conclusions.

7.2 Clinical Safety Controls

- Predefined abstention and escalation states when evidence is insufficient, conflicted, corrupted or out of distribution.
- Independent confirmation requirements for high-consequence outputs.
- Explicit limitations, contraindications and intended-use boundaries.
- Safety-related interface requirements and human-factors validation.
- Subgroup, site, device and acquisition-environment evaluation.
- Post-deployment safety monitoring, incident triage, rollback and field communication as applicable.

8

Software Lifecycle and Configuration Governance

8. Software Lifecycle and Configuration Governance

[IEC 62304](#) defines lifecycle processes for medical-device software. The FDA's [Content of Premarket Submissions for Device Software Functions](#) is also a relevant documentation reference for software functions that meet the device definition. xBxBio's next phase should translate these principles into controlled, risk-proportionate software lifecycle records.

8.1 Minimum Software Governance Controls

- Controlled software-development plan and lifecycle model.

- Architecture, interfaces, data flows, trust boundaries and third-party dependencies.
- System, software, data, model, cybersecurity and usability requirements.
- Configuration identifiers for code, infrastructure, data schemas, prompts, models, parameters and dependencies.
- Code review, static analysis, dependency scanning, test automation and release approval.
- Anomaly, defect, vulnerability and technical-debt management.
- Software bill of materials and component provenance where applicable.
- Environment segregation for development, test, validation and production.
- Reproducible builds, deployment records, rollback and disaster-recovery controls.
- Verification evidence traceable to requirements and risk controls.

9

Data Integrity, Provenance and Interoperability

9. Data Integrity, Provenance and Interoperability

Connected cardiovascular intelligence depends on more than collecting data. It requires preservation of identity, chronology, semantics, units, acquisition conditions, transformation history, quality, permissions, and clinical context across heterogeneous systems and modalities.

9.1 Source-to-Inference Lineage

Every derived information state should be reconstructable from its source data and processing history. At minimum, lineage should identify the patient or research subject identity domain, source system, acquisition device, encounter, timestamp, time zone, author or device, units, transformation, data-quality state, software and model version, parameterization, threshold, and validation status.

9.2 Interoperability

Standards such as [HL7 FHIR](#) and [DICOM](#) may support exchange and representation of health and imaging information. Standards-based transport does not, by itself, guarantee semantic correctness, temporal alignment, patient matching, data quality, clinical relevance, or fitness for a specific inference.

9.3 Data-Quality States

- Available and conformant
- Available with known limitation
- Missing by design
- Unexpectedly missing
- Conflicted across sources
- Out of range or physiologically implausible
- Temporally misaligned
- Identity uncertain
- Transformation not reproducible
- Provenance incomplete

10

Clinical AI Governance and Defensible Information States

10. Clinical AI Governance and Defensible Information States

The FDA's [Good Machine Learning Practice principles](#), [transparency principles for machine-learning-enabled medical devices](#), and [AI-enabled device lifecycle draft guidance](#) emphasize lifecycle, data, human-AI interaction, transparency, performance and risk considerations. xBxBio's framework incorporates these subjects as governance domains without asserting current regulatory conformity.

10.1 Defensible Information States

Information state	Meaning
Evidence-sufficient	Inputs, provenance, quality, applicability, calibration and validation support the defined output within its intended use.
Evidence-limited	Some evidence is available, but missingness, uncertainty, scope or validation limitations materially constrain interpretation.
Evidence-conflicted	Sources disagree in a way that cannot be resolved by predefined rules or additional context.
Out-of-distribution	The case, device, site, population, acquisition condition or data pattern falls outside the supported validation domain.
Abstention	The system intentionally withholds an inference because preconditions for a defensible output are not met.
Escalation	The system directs the matter to an accountable qualified human or defined workflow because risk, uncertainty or conflict exceeds a threshold.
Invalid or unavailable	Data integrity, system state, cybersecurity, configuration or service availability prevents safe use.

10.2 Minimum AI Governance Controls

- Model purpose, intended output, prohibited use and clinical consequence.
- Training, tuning, test and external-evaluation dataset boundaries.
- Data representativeness, labeling, reference standards and subgroup analysis.
- Performance metrics, calibration, uncertainty and decision thresholds.
- Out-of-distribution detection, drift surveillance and trigger thresholds.
- Human review, override, escalation, audit trail and accountability.
- Model, prompt, retrieval, tool and parameter versioning where applicable.
- Change impact, re-verification, revalidation, rollback and communication.

11

Cardiovascular State Estimation and Computational Physiology

11. Cardiovascular State Estimation and Computational Physiology

Technical positioning

The strongest future xBxBio architecture is not a collection of isolated classifiers. It is a provenance-preserving, longitudinal cardiovascular state-estimation environment in which multimodal observations update bounded representations of evolving electrophysiologic, biomechanical, vascular, metabolic and hemodynamic states.

11.1 Technical Governance Statement

Yes—but only if the platform is engineered and validated as a [constrained Bayesian cardiovascular state-estimation system](#), not a conventional classifier. [ECG and electrophysiologic inverse problems](#), echocardiographic and CT/MR phenotypes, invasive and noninvasive hemodynamics, laboratory biomarkers, genomics, pharmacotherapy, interventions, and longitudinal treatment history may contribute to estimates of evolving electrophysiologic, biomechanical, vascular, and hemodynamic states through [hierarchical state-space inference](#). [Computational fluid dynamics](#) and [fluid-solid interaction modeling](#) may impose physiologic constraints on pressure-flow coupling, myocardial wall stress, ventricular-arterial interaction, chamber mechanics, and perfusion, while appropriately designed [causal models](#) may assist in distinguishing treatment-associated effects from measured confounding.

Clinically, posterior probability alone is insufficient. Uncertainty must propagate across modalities; missingness, acquisition variability, [distributional shift and out-of-distribution conditions](#) must be characterized; and predefined [abstention and escalation states](#) must prevent unsupported inferences from being represented as clinically actionable conclusions. Every derived information state must remain traceable to its source data, acquisition context, transformation history, software and model version, validation status, assumptions, and applicable risk controls. Under [FDA Quality Management System Regulation \(QMSR\)](#) and design-control principles, together with relevant considerations from [ISO 13485](#), [ISO 14971](#), [ISO/TS 24971-2](#), [IEC 62304](#), and [IEC 62366-1](#), the real test is whether the system is physiologically coherent, calibrated, reproducible, traceable, auditable, appropriately interpretable, risk-controlled, and lifecycle-valid for its defined intended use—not merely statistically accurate.

Current-status qualification

These concepts describe xBxBio's research and governance direction. They do not represent claims that the production platform currently performs these functions in a clinically validated, externally evaluated, deployed, or regulator-authorized environment.

11.2 Conceptual State-Space Formulation

A future implementation may represent a latent cardiovascular state x_t that evolves through time under prior state, interventions u_t , patient-specific parameters, and process uncertainty. Each modality m produces observations $y_t^{(m)}$ conditional on the latent state and modality-specific quality $q_t^{(m)}$. The system would update a posterior state estimate as new evidence arrives while preserving the contribution, timing and uncertainty of each observation.

State transition: $x_t \sim p(x_t | x_{(t-1)}, u_t, \theta)$

Observation model: $y_t^{(m)} \sim p(y_t^{(m)} | x_t, q_t^{(m)})$

Bayesian update: $p(x_t | y_{(1:t)})$ is proportional to $p(y_t | x_t) \times p(x_t | y_{(1:t-1)})$

This formulation is conceptual. It does not prescribe a specific algorithm, model family, inference engine, physiological solver, or clinical claim. Any implementation would require problem-specific identifiability analysis, observability analysis, numerical stability evaluation, data-quality controls, parameter estimation, uncertainty characterization, validation, and clinical interpretation boundaries.

11.3 Technical Background Resources

[Bayesian inference for fitting cardiac models to experimental data](#) — PubMed.

[Hybrid neural state-space modeling for electrocardiographic imaging](#) — PubMed.

[Electrocardiographic inverse-problem methods and validation review](#) — PubMed.

[Computational fluid-dynamics modeling in cardiovascular medicine review](#) — PubMed.

[Fluid-structure interaction modeling in cardiovascular medicine review](#) — PubMed.

[Causal inference about intervention effects from observational data](#) — PubMed.

12

Evidence and Validation Architecture

12. Evidence and Validation Architecture

Validation is not a single final test and cannot be reduced to one aggregate performance metric. The required evidence depends on intended use, clinical consequence, system boundaries, modality, workflow, population, site, acquisition environment, degree of automation, and the claims that xBxBio proposes to make.

Evidence domain	Question addressed
Scientific validity	Whether the clinical, physiological or analytical association underlying a function is sufficiently supported.
Analytical and technical validation	Whether acquisition, ingestion, transformation, calculation and output generation perform as specified.
Software verification	Whether implemented software outputs satisfy specified design inputs and technical requirements.
Clinical validation	Whether performance is clinically meaningful for the defined intended use, population and context.
Generalizability and robustness	Whether performance remains acceptable across relevant sites, devices, populations, subgroups and acquisition conditions.
Calibration and uncertainty	Whether probabilities or scores correspond to observed outcomes and uncertainty is appropriately characterized.
Human-factors validation	Whether intended users can correctly perceive, understand and act on information under realistic use conditions.
Workflow and decision-impact evaluation	Whether the function improves the intended process or decision without introducing unacceptable new risk.
Lifecycle validation	Whether performance, safety and controls remain acceptable following relevant change and over time.

12.1 Minimum Claim-Specific Evidence Record

- Exact claim and intended audience.
- Platform, software, model, data and configuration version.
- Target population and clinically relevant subgroups.
- Study design, sites, inclusion/exclusion criteria and sample disposition.
- Reference standard, endpoint definitions and adjudication process.
- Prespecified statistical analysis plan and acceptance criteria.
- Sensitivity, specificity, predictive values, calibration and confidence intervals as relevant.
- Missing-data handling, bias assessment, multiplicity and sensitivity analyses.
- Robustness across devices, sites, acquisition protocols and time.
- Failure analysis, limitations, uncertainty and residual risk.
- Independent review and approval for public communication.

12.2 Performance Metrics Beyond AUROC

For clinical or longitudinal cardiovascular functions, a high area under the receiver operating characteristic curve may still coexist with poor calibration, low positive predictive value, clinically unacceptable false negatives, subgroup failure, temporal instability, or little decision benefit. Metrics should therefore be selected according to the intended use and may include calibration slope/intercept, Brier score, precision-recall performance, time-dependent discrimination, decision-curve analysis, net benefit, sensitivity at a clinically defined specificity, subgroup error rates, alert burden, time-to-event performance, and failure-to-abstain rate.

13

Human Factors and Responsible Clinical Use

13. Human Factors and Responsible Clinical Use

The FDA's current [Applying Human Factors and Usability Engineering to Medical Devices guidance](#) and [IEC 62366-1](#) provide relevant reference points for minimizing use-related risk. Human oversight must be implemented through the interface and workflow, not merely asserted in a disclaimer.

13.1 Human-Factors Questions

- Can the intended user identify the data and assumptions underlying the output?
- Can the user distinguish observation, inference, uncertainty, limitation and recommendation?
- Does the interface make conflicting or missing evidence visible?
- Are abstention and escalation states understandable and actionable?
- Could color, ranking, urgency, defaults or automation create inappropriate trust or dismissal?
- Can a user recover from error, override an output and document the rationale?
- Does the workflow support accountable review without creating excessive burden or alert fatigue?
- Are training, labeling, help content and on-screen information consistent with the validated use environment?

13.2 Responsible Clinical-Use Boundary

Until an appropriate production configuration is developed, verified, validated, externally evaluated as required, and authorized where applicable, xBxBio public materials should not represent the platform as diagnosing disease, prescribing treatment, selecting therapy, replacing a clinician, or providing patient-specific clinical advice.

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Cybersecurity, Privacy and Operational Resilience

14. Cybersecurity, Privacy and Operational Resilience

The FDA's [February 2026 cybersecurity guidance](#) addresses cybersecurity device design, quality-system considerations and recommended premarket documentation for devices with cybersecurity risk. xBxBio should incorporate security and resilience into architecture, risk management, software lifecycle and supplier control from the beginning.

14.1 Minimum Security and Resilience Domains

- Security governance, asset ownership, threat modeling and risk acceptance.
- Identity, authentication, authorization, least privilege and privileged-access control.
- Encryption, key management, secrets management and certificate lifecycle.
- Secure development, code review, dependency management and vulnerability testing.

- Cloud configuration, network segmentation, logging, monitoring and tamper resistance.
- Software bill of materials, component provenance and supplier vulnerability response.
- Backup, restoration, continuity, disaster recovery and validated rollback.
- Incident detection, triage, containment, notification, remediation and lessons learned.
- Privacy, data minimization, purpose limitation, retention, deletion and lawful data use.
- Protection against model extraction, prompt injection, data poisoning, adversarial manipulation and tool misuse where relevant.

No current security certification claim

This framework does not establish that xBxBio has completed production security implementation, penetration testing, SOC 2 examination, ISO/IEC 27001 certification, HIPAA assessment, or any other external security or privacy attestation.

15

Lifecycle Monitoring and Controlled Change

15. Lifecycle Monitoring and Controlled Change

A production cardiovascular intelligence system can become unsafe even if its original release was adequately validated. Clinical practice, populations, devices, data sources, software dependencies, cybersecurity threats, infrastructure, workflows and model behavior change over time. The framework therefore treats monitoring and controlled change as design requirements.

For AI-enabled device functions, the FDA's final [Predetermined Change Control Plan guidance](#) is a relevant source when an authorized PCCP is applicable to the future device and modification strategy. xBxBio must not imply that a PCCP exists or has been authorized unless that actually occurs.

15.1 Monitoring Domains

- Input data quality, completeness, source changes and semantic drift.
- Population, site, device, workflow and acquisition-condition shift.
- Performance, calibration, subgroup behavior and failure-to-abstain rate.
- User interaction, overrides, alert burden, workarounds and use errors.
- Software anomalies, latency, availability, dependencies and infrastructure health.
- Cybersecurity vulnerabilities, threats, incidents and supplier advisories.
- Clinical safety signals, complaints, incidents and emerging evidence.
- Model, prompt, retrieval, threshold, code, data-schema and configuration changes.

15.2 Change-Control Minimums

- Change description and rationale.
- Affected intended use, claims, users, interfaces, risks and evidence.
- Regulatory-impact assessment.
- Cybersecurity, privacy and interoperability impact.
- Required verification, validation, human-factors or clinical evidence.
- Approval authority, release conditions, monitoring and rollback.
- Communication to users, partners, regulators or customers when applicable.

16

Controlled Implementation and Validation-Readiness Plan

16. Controlled Implementation and Validation-Readiness Plan

The next phase is not a press-release concept. It is a controlled work program that converts the approved framework into actionable requirements, records, responsibilities and evidence plans. The following workstreams represent the minimum implementation sequence.

Workstream	Required outcome
1. Intended use and claims	Define product/function boundaries, users, context, clinical consequence, limitations and prohibited use.
2. Regulatory strategy	Perform jurisdiction- and function-specific analysis; document assumptions and engagement strategy.
3. Quality-system implementation	Establish controlled documents, records, training, management responsibility, design controls and supplier controls.
4. Architecture and traceability	Baseline system architecture, interfaces, data flows, requirements and source-to-output traceability.
5. Risk and clinical safety	Create hazard analysis, risk controls, safety requirements, residual-risk criteria and safety-review process.
6. Data governance and provenance	Define data rights, identity, quality, lineage, semantics, retention, access and acceptable-use controls.
7. Software and model governance	Establish lifecycle plans, configuration, development controls, evaluation, uncertainty, drift and change management.
8. Cybersecurity and privacy	Threat model the intended deployment; implement security, privacy, resilience and incident-response controls.
9. Verification and validation master planning	Define test strategy, acceptance criteria, environments, independence and evidence packages.
10. Human factors and workflow	Identify users, critical tasks, use-related hazards, interface requirements, formative work and validation needs.
11. External collaboration readiness	Prepare data-use, research, institutional, security, quality and publication requirements for collaborators.
12. Public-communications governance	Align website, LinkedIn, PRLog, Crunchbase, GitHub, Zenodo and investor materials with approved evidence.

16.1 Minimum Next-Phase Deliverables

- Approved intended-use and claims baseline.
- Regulatory strategy memorandum and open-question register.
- Quality-system implementation plan and controlled-document hierarchy.
- System architecture baseline and master traceability strategy.
- Risk-management plan and initial clinical safety risk analysis.
- Software-development and configuration-management plans.
- Data-governance, provenance and interoperability specifications.
- AI/model governance plan and evaluation framework.
- Cybersecurity and privacy plans, threat model and resilience requirements.
- Verification and validation master plan.
- Human-factors engineering plan.
- External clinical/research collaboration package.
- Public claims registry and communications approval process.

17. Funding and External Collaboration Dependencies

Internal framework approval strengthens xBxBio's readiness, but it does not remove the need for capital, qualified personnel, infrastructure, data access, external collaborators and independent evidence. The transition from governance foundation to validated deployment is resource-intensive by design because each high-consequence claim must be implemented, tested and supported.

17.1 Financing Would Enable

- Production-grade cloud, security, observability, backup, recovery and operations infrastructure.
- Controlled software, data, AI/model, quality, regulatory, clinical-safety and cybersecurity staffing.
- Licenses, development environments, test environments, validation environments and qualified tools.
- Lawful access to representative cardiovascular data and appropriate reference standards.
- Clinical, academic, health-system and medical-device collaboration.
- Verification, validation, usability, cybersecurity and independent evaluation.
- Regulatory consultation and submissions where applicable.
- Insurance, legal, privacy, contracting, supplier and institutional-review infrastructure.

17.2 External Collaboration Priorities

- Cardiovascular institutions with multimodal longitudinal data and qualified clinical investigators.
- Academic groups with expertise in electrophysiology, imaging, hemodynamics, computational physiology, biostatistics and causal inference.
- Medical-device and diagnostic organizations able to support device, data and workflow integration.
- Cloud, cybersecurity, interoperability and data-engineering organizations willing to operate under controlled requirements.
- Regulatory, quality, human-factors and clinical-evaluation specialists with relevant device-software experience.

Funding statement for public use

The next phase requires financing to translate xBxBio's internally approved governance framework into production infrastructure, controlled implementation, verification and validation, external evaluation, and any applicable regulatory activities. This is an execution requirement—not evidence that the platform is currently available or validated.

18

Public Claims and Communications Control

18. Public Claims and Communications Control

Public communication is part of the control environment. Inconsistent or overstated language across xBxBio.com, LinkedIn, PRLog, Crunchbase, GitHub, Zenodo, investor materials, presentations or email can undermine due diligence and create regulatory, scientific and reputational risk.

Topic	Permitted wording now	Do not state now
Framework status	"Internally reviewed and approved."	"Regulator approved," "FDA approved," or "certified."

Topic	Permitted wording now	Do not state now
Current phase	"Advancing into controlled implementation and validation readiness."	"Ready for clinical deployment."
Platform status	"In pre-commercial research and engineering development."	"Available for diagnosis or treatment."
Validation	"Validation planning and evidence development are future phases."	"Clinically validated" without completed evidence.
Performance	No numerical claim unless supported by an approved evidence record.	Sensitivity, specificity, accuracy, savings or outcome claims without substantiation.
Partnerships	"Seeking" or "in discussion with" when accurate and approved.	Named partnership, endorsement or validation without written authorization.
Standards	"Framework references" or "intended alignment" when precise.	"ISO compliant," "QMSR compliant," or "fully validated" without evidence.
Patent	"Patent pending" when the filing status remains accurate.	"Patented" or "patent granted" unless granted.

18.1 Approved Core Announcement Language

Exact wording

xBxBio has completed the internal review and approval of its Regulatory-by-Design Framework for Connected Cardiovascular Intelligence. Internal review and approval are complete. xBxBio is now advancing from framework definition into controlled implementation and validation readiness.

Exact clarification

This represents an internal xBxBio governance milestone. It does not constitute FDA clearance, FDA approval, regulatory authorization, certification, conformity assessment, or endorsement by any external regulatory authority or standards organization.

19. Development Status and Mandatory Disclosures

19.1 Full Development-Status Statement

Exact text for public use

The xBxBio Connected Cardiovascular Intelligence Platform is in pre-commercial research and engineering development. Selected non-proprietary scientific, technical and governance materials are publicly available; however, the production platform is not currently offered for clinical diagnosis, treatment, patient management or commercial deployment. xBxBio has not received FDA clearance, FDA approval, regulatory authorization, certification or conformity assessment for the platform. Future clinical or commercial availability will depend on completion of production development, verification and validation, cybersecurity and quality controls, external evaluation, financing, and any applicable regulatory pathway.

19.2 Short Footer Statement

Exact text for website and public materials

The xBxBio platform is under development and is not currently cleared, approved, authorized or offered for clinical use. Materials are provided for scientific, technical, corporate and educational purposes only and do not constitute medical advice.

19.3 No Medical Advice

Nothing in this document should be used to diagnose, treat, prevent, monitor or manage a medical condition. Individuals should seek advice from a qualified healthcare professional for patient-specific questions or care.

20

Conclusion

20. Conclusion

Completion of internal review and approval of the xBxBio Regulatory-by-Design Framework is a meaningful governance milestone because it establishes the principles by which future development should be controlled. It does not substitute for implementation, evidence, independent evaluation, regulatory authorization, financing or deployment.

The immediate objective is to convert the framework into controlled requirements, accountable roles, traceable records, risk controls, validation plans, security requirements, human-factors activities, and lifecycle monitoring. The quality of that transition will determine whether xBxBio's technical ambition can become a reproducible, auditable and clinically defensible system.

Development sequence

Architecture → Internal Governance Approval → Controlled Implementation → Evidence Development → Verification and Validation → External Evaluation → Applicable Regulatory Pathway → Deployment

xBxBio's public position is therefore clear: the governance foundation has been approved internally; controlled implementation and validation readiness begin next; and production deployment remains contingent on funding, execution, evidence, external collaboration, and applicable regulatory requirements.

APPENDIX A

Terminology

Appendix A. Terminology

Term	Definition
Abstention	A predefined state in which the system intentionally withholds an inference or recommendation because conditions for a defensible output are not satisfied.
Analytical validation	Evidence that data acquisition, processing, transformation, calculation or measurement performs as specified.
Calibration	Agreement between predicted probabilities or risks and observed outcomes within the defined context.
Clinical validation	Evidence that a function achieves clinically meaningful performance for its defined intended use, population and context.
Configuration	The identifiable combination of software, infrastructure, data schema, model, parameters, dependencies, thresholds and settings that produces an output.
Controlled implementation	Translation of approved framework principles into effective requirements, procedures, records, roles, controls and evidence plans.
Defensible information state	An information state whose source, context, quality, assumptions, uncertainty, applicability, validation and limitations are sufficiently characterized for its defined use.
Distributional shift	A material change in the statistical or clinical characteristics of data relative to the domain used for development or validation.
Drift	Change over time in input data, model behavior, calibration, performance, workflow or outcome relationships.
External evaluation	Assessment by qualified parties outside the internal development team, which may include independent retrospective or prospective evaluation.
Human oversight	Operationally defined review, decision, override, escalation and accountability by qualified people; not merely a disclaimer.
Intended use	The objective intent regarding the use of a product or function, including users, population, context, inputs, outputs and clinical role.
Out-of-distribution	A case or data pattern materially outside the supported development or validation domain.
Provenance	The origin, custody, timing, acquisition context, transformation history, version and processing lineage of information.
Regulatory-by-Design	A development approach in which intended use, risk, quality, evidence, human factors, cybersecurity, traceability and lifecycle considerations are incorporated before deployment.
Validation readiness	A state in which intended use, requirements, risks, configuration, acceptance criteria, environments, data and protocols are sufficiently controlled to begin formal validation.

APPENDIX B

Linked Authorities and Technical Resources

Appendix B. Linked Authorities and Technical Resources

All links below were selected from official regulatory, standards, interoperability, scientific-indexing, or xBxBio sources and were reviewed for use in this publication package on September 2, 2026. External organizations may change URLs after publication; xBxBio should perform a final click-through check immediately before the September 17 release.

1. FDA — Quality Management System Regulation (QMSR)

Current FDA QMSR information, including the February 2, 2026 effective date and incorporation by reference of ISO 13485:2016. <https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr>

2. FDA — Guidances with Digital Health Content

Current index of FDA digital-health guidance documents. <https://www.fda.gov/medical-devices/digital-health-center-excellence/guidances-digital-health-content>

3. FDA — Clinical Decision Support Software

Final FDA guidance on clinical decision support software. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/clinical-decision-support-software>

4. FDA — Good Machine Learning Practice for Medical Device Development

Current GMLP principles and link to the IMDRF final document. <https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles>

5. FDA — AI-Enabled Device Software Functions: Lifecycle Management and Marketing Submission Recommendations

Draft FDA guidance addressing AI-enabled device lifecycle and submission considerations. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/artificial-intelligence-enabled-device-software-functions-lifecycle-management-and-marketing>

6. FDA — Predetermined Change Control Plan for AI-Enabled Device Software Functions

Final FDA guidance issued August 2025. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/marketing-submission-recommendations-predetermined-change-control-plan-artificial-intelligence>

7. FDA — Cybersecurity in Medical Devices

Final FDA cybersecurity guidance dated February 2026. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/cybersecurity-medical-devices-quality-management-system-considerations-and-content-premarket>

8. FDA — Applying Human Factors and Usability Engineering to Medical Devices

Final FDA human-factors guidance dated August 2026. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/applying-human-factors-and-usability-engineering-medical-devices>

9. FDA — Content of Premarket Submissions for Device Software Functions

Final FDA guidance describing recommended software documentation. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/content-premarket-submissions-device-software-functions>

10. FDA — Transparency for Machine Learning-Enabled Medical Devices

Joint transparency guiding principles. <https://www.fda.gov/medical-devices/software-medical-device-samd/transparency-machine-learning-enabled-medical-devices-guiding-principles>

11. ISO 13485:2016 — Medical devices — Quality management systems

Official ISO page. <https://www.iso.org/standard/59752.html>

12. ISO 14971:2019 — Medical devices — Application of risk management

Official ISO page. <https://www.iso.org/standard/72704.html>

13. ISO/TR 24971:2020 — Guidance on ISO 14971

Official ISO page. <https://www.iso.org/standard/74437.html>

14. ISO/TS 24971-2:2026 — Machine learning in artificial intelligence

Official ISO page, including the stated scope limitation for LLM/generative AI. <https://www.iso.org/standard/24971-2>

15. IEC 62304 — Medical device software — Software life cycle processes

Official IEC webstore page. <https://webstore.iec.ch/en/publication/22794>

16. IEC 62366-1:2015 — Application of usability engineering to medical devices

Official ISO page for the jointly published standard. <https://www.iso.org/standard/63179.html>

17. ISO/IEC 23894:2023 — Artificial intelligence — Guidance on risk management

Official ISO page. <https://www.iso.org/standard/77304.html>

18. ISO/IEC 42001:2023 — Artificial intelligence management systems

Official ISO page. <https://www.iso.org/standard/42001>

19. PubMed — Bayesian Inference for Fitting Cardiac Models to Experiments

Peer-reviewed cardiac-modeling methods source indexed by the U.S. National Library of Medicine. <https://pubmed.ncbi.nlm.nih.gov/36322242/>

20. PubMed — Hybrid Neural State-Space Modeling for Electrocardiographic Imaging

Peer-reviewed state-space modeling source indexed by the U.S. National Library of Medicine. <https://pubmed.ncbi.nlm.nih.gov/38478452/>

21. PubMed — Solving the Inverse Problem of Electrocardiography

Peer-reviewed review indexed by the U.S. National Library of Medicine. <https://pubmed.ncbi.nlm.nih.gov/39453795/>

22. PubMed — Computational Fluid Dynamics Modelling in Cardiovascular Medicine

Peer-reviewed review indexed by the U.S. National Library of Medicine. <https://pubmed.ncbi.nlm.nih.gov/26512019/>

23. PubMed — Fluid-Structure Interaction Modeling in Cardiovascular Medicine

Peer-reviewed review indexed by the U.S. National Library of Medicine. <https://pubmed.ncbi.nlm.nih.gov/32081559/>

24. PubMed — Causal Inference About Effects of Interventions From Observational Data

Peer-reviewed methodological source indexed by the U.S. National Library of Medicine. <https://pubmed.ncbi.nlm.nih.gov/38722735/>

25. HL7 FHIR

Official HL7 FHIR specification site. <https://www.hl7.org/fhir/>

26. DICOM Standard

Official DICOM standard publication site. <https://dicom.nema.org/medical/dicom/current/output/html/part01.html>

27. xBxBio Official Website

Primary xBxBio corporate website. <https://xbxbio.com/>

28. xBxBio LinkedIn Company Page

Official company page. <https://www.linkedin.com/company/xbx-bio/>

29. xBxBio Connected Cardiovascular Intelligence Repository

Public GitHub research repository. <https://github.com/xbxbio/connected-cardiovascular-intelligence>

30. xBxBio Public Research Site

Public GitHub Pages site. <https://xbxbio.github.io/connected-cardiovascular-intelligence/>

31. xBxBio PRLog Newsroom

External xBxBio press-release archive. <https://pressroom.prlog.org/xBxBio/>

APPENDIX C

Publication Metadata

Appendix C. Publication Metadata

C.1 Zenodo Metadata

Field	Exact value
Resource type	Publication — Report
Title	xBxBio Regulatory-by-Design Framework for Connected Cardiovascular Intelligence
Subtitle	Public Governance Summary Following Internal Review and Approval
Version	1.0
Publication date	2026-09-17
Creator	xBxBio
Publisher	xBxBio
Language	English
Rights	All rights reserved — xBxBio
DOI	10.5281/zenodo.22257444

C.2 Zenodo Description — Exact Copy

Copy and paste into the Zenodo Description field

xBxBio has completed the internal review and approval of its Regulatory-by-Design Framework for Connected Cardiovascular Intelligence. With internal review and approval complete, xBxBio is advancing from framework definition into controlled implementation planning, evidence development, verification and validation readiness, and lifecycle governance. This public governance summary describes selected non-proprietary principles addressing intended use, quality and design controls, software lifecycle governance, risk management, data integrity, provenance, clinical AI governance, uncertainty, human oversight, cybersecurity, verification and validation, change control, and lifecycle monitoring. The milestone represents an internal xBxBio governance decision. It does not constitute FDA clearance, FDA approval, regulatory authorization, certification, conformity assessment, or endorsement of the xBxBio platform by any external organization. The platform remains in pre-commercial research and engineering development and is not currently offered for clinical use or commercial deployment.

C.3 Keywords

Connected Cardiovascular Intelligence; cardiovascular AI; clinical AI governance; regulatory science; quality management; risk management; data integrity; provenance; validation; human oversight; lifecycle monitoring; digital health; cardiovascular informatics; software as a medical device; multimodal cardiovascular data.

C.4 GitHub Release Metadata

Field	Exact value
Repository	xbxbio/connected-cardiovascular-intelligence
Folder	publications/XBX-RSP-RF-001/
Tag	XBX-RSP-RF-001-v1.0
Release title	XBX-RSP-RF-001 — xBxBio Regulatory-by-Design Framework for Connected Cardiovascular Intelligence
PDF filename	XBX-RSP-RF-001_xBxBio_Regulatory-by-Design_Framework_Public_Governance_Summary_v1.0.pdf
Word filename	XBX-RSP-RF-001_xBxBio_Regulatory-by-Design_Framework_Public_Governance_Summary_v1.0.docx

C.5 GitHub Release Notes — Exact Copy

Copy and paste into GitHub Release Notes

xBxBio has published a public governance summary following completion of the internal review and approval of its Regulatory-by-Design Framework for Connected Cardiovascular Intelligence. The milestone advances xBxBio from framework definition into controlled implementation planning, evidence development, verification and validation readiness, and lifecycle governance. This is an internal xBxBio governance milestone and does not represent FDA clearance, FDA approval, regulatory authorization, certification, conformity assessment, or external regulatory endorsement. DOI: [10.5281/zenodo.22257444](https://doi.org/10.5281/zenodo.22257444). Official Website: <https://xbxbio.com>. LinkedIn: <https://www.linkedin.com/company/xbx-bio/>.



Connected Cardiovascular Intelligence™

Publication status

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Official Website: <https://xbxbio.com>

LinkedIn: <https://www.linkedin.com/company/xbx-bio/>

Public Research Repository: <https://github.com/xbxbio/connected-cardiovascular-intelligence>

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